Hyperbiliverdinemia (Green Jaundice): A Comprehensive Disease Profile

Disease: Hyperbiliverdinemia (green jaundice) · **MONDO:** MONDO:0013595 · **OMIM:** 614156
Causal gene: BLVRA (biliverdin reductase A; HGNC:1062; NCBI Gene 644; gene MIM *109750; UniProt P53004; chr7p13, GRCh38 chr7:43,758,009–43,808,868 (+ strand); ENSG00000106605; canonical transcript ENST00000265523) · Category:** Mendelian, autosomal recessive

Summary

Hyperbiliverdinemia, also called **green jaundice**, is a rare autosomal-recessive inborn error of heme catabolism caused by **biallelic loss-of-function variants in *BLVRA***, the gene encoding **biliverdin reductase A (BVRα; EC 1.3.1.24)**. Under normal physiology, heme is degraded by heme oxygenase to the green pigment **biliverdin IXα**, which BVRα then reduces, using NADPH/NADH, to the yellow pigment **bilirubin IXα**. When BVRα is absent or non-functional, this final reduction step fails and biliverdin IXα accumulates in serum, urine, bile and milk, producing a characteristic green discoloration of body fluids and tissues (PMID: 21278388, PMID: 27387768).

The disease was defined molecularly by Nytofte and colleagues in 2011, who identified a **homozygous nonsense mutation c.214C>A (p.Ser44*)** in *BLVRA* in two unrelated Inuit women from Greenland. Strikingly, the clinical phenotype — green jaundice — is **latent**: it becomes visible only when a superimposed **cholestasis / biliary obstruction** raises the metabolic flux through the heme-degradation pathway, and it resolves after the obstruction is surgically relieved (PMID: 21278388). The causal allele is a **population-restricted founder variant** enriched in the Greenlandic Inuit, where it reaches ~1% homozygosity and ~4.5% heterozygosity (PMID: 27387768).

The condition is **biochemically benign**. *BLVRA* is **loss-of-function tolerant** in population databases (gnomAD pLI ≈ 0, observed/expected LoF = 0.67), heterozygous carriers are asymptomatic, and even homozygous individuals are healthy in the absence of cholestasis — the index family included unaffected homozygous siblings, indicating **reduced penetrance**. Management is directed at the **underlying cholestatic process**, not at the enzyme defect.

Beyond its namesake reductase activity, BVRα is a well-characterized **"moonlighting" protein** acting as a Ser/Thr/Tyr kinase, scaffold, and transcription factor in insulin, MAPK/ERK and PKC signaling; interestingly, the complete human BVRα null is clinically mild, suggesting these non-canonical functions are either dispensable or redundant in humans.

Key Findings

Finding 1 — Hyperbiliverdinemia is caused by biallelic loss-of-function *BLVRA* variants

The defining report (Nytofte et al., 2011) studied two unrelated Inuit women from different geographical areas of Greenland who exhibited episodic green jaundice during biliary obstruction. Coding-sequence analysis of the biliverdin reductase alpha gene (*BLVRA*) detected three single-nucleotide changes — c.90G→A, c.214C→A and c.743A→C — predicting p.Ala3Thr, **p.Ser44X** and p.Gly220Gly, of which the **c.214C→A (p.Ser44X) nonsense allele** is the pathogenic driver (PMID: 21278388). HPLC–MS/MS demonstrated **hypercholanaemia and high concentrations of biliverdin IXα in serum, urine, bile and milk**, with absent or only moderate hyperbilirubinemia; crucially, the **hyperbiliverdinaemia disappeared after surgical correction of the cholestasis**.

> *"Analysis of the coding sequence of the biliverdin reductase alpha (BVRα) gene (BLVRA) detected three single-nucleotide polymorphisms: c.90G→A, c.214C→A and c.743A→C, which result in p.Ala3Thr, p.Ser44X and p.Gly220Gly"* — PMID: 21278388
>
> *"high-performance liquid chromatography tandem mass spectrometry showed hypercholanaemia and high concentrations of biliverdin IXα in serum, urine, bile and milk. Hyperbiliverdinaemia disappeared after surgical correction of the cholestasis"* — PMID: 21278388

The segregation pattern supported autosomal-recessive inheritance with **reduced penetrance**: both parents of one patient were heterozygous, while a homozygous brother and the proband's homozygous sister were **unaffected** — consistent with the requirement for a second, cholestatic "hit" to unmask the phenotype. Functional confirmation came from in-vitro expression work showing the variant yields a dead enzyme:

> *"The mutation (c.214C>A) in BLVRA results in a non-functional enzyme (mutBVRα). Consequently, homozygous carriers suffering from cholestasis develop green jaundice."* — PMID: 27387768

Evidence type: human clinical + in-vitro functional. **Confidence:** High.

Finding 2 — The p.Ser44* allele is a Greenlandic Inuit founder variant

Gonzalez-Sanchez et al. (2016) screened 311 individuals in Greenland and found the mutant *BLVRA* allele at appreciable frequency — approximately **1% homozygous (a/a) and 4.5% heterozygous (A/a)**. Together with the observation that the two index cases came from different geographical areas of Greenland, this is consistent with a **population-specific founder allele** rather than recurrent mutation.

> *"a screening of the mut-BLVRA allele (a) in 311 individuals in Greenland revealed that this SNP was relatively frequent in the Inuit population studied (1% a/a and 4.5% A/a)"* — PMID: 27387768

Evidence type: human population genetics. **Confidence:** High (single population).

Finding 3 — BVRα is a moonlighting protein (reductase + kinase + scaffold + transcription factor)

Human biliverdin reductase (hBVR / BVRα, EC 1.3.1.24) is a dual-cofactor (NADPH/NADH) oxidoreductase that also moonlights as a **Ser/Thr/Tyr kinase, scaffold/anchor protein, and bZip transcription factor**. It acts as an **upstream activator of the insulin/IGF-1 signaling pathway** and of multiple **PKC isoforms**, is required for ERK1/2 activation and nucleocytoplasmic shuttling, and forms an **hBVR–PKCδ–ERK2 ternary complex** essential for ERK2-mediated gene activation (PMID: 22419908, PMID: 22065579, PMID: 17919068).

> *"hBVR is a Ser/Thr/Tyr-kinase, a scaffold protein, a transcription factor, and an intracellular transporter of gene regulators. hBVR is an upstream activator of the insulin/IGF-1 signaling pathway and of protein kinase C (PKC) kinases"* — PMID: 22419908

Mouse genetics confirm that catalytic and non-catalytic roles are **separable**: catalytic-dead *Blvra* missense mutants (G17A, E97A) retain non-canonical functions while losing ~80–95% of oxidoreductase activity (see Finding 5). The clinical mildness of the human p.Ser44* null — which removes **all** of these functions — indicates that, in humans, loss of BVRα's moonlighting roles is tolerated, at least under normal environmental conditions.

Evidence type: in-vitro biochemistry + model organism. **Confidence:** High for biochemistry; interpretation of human tolerance is inferred.

Finding 4 — *BLVRA* is loss-of-function tolerant; the Inuit allele is essentially absent from gnomAD

gnomAD v4 constraint metrics for *BLVRA* (ENSG00000106605; canonical transcript ENST00000265523; chr7p13, ~43.78–43.81 Mb, GRCh38) show the gene is **not haploinsufficient**: pLI = 1.2×10^{-5}, observed/expected LoF = **0.67** (90% CI 0.48–0.96), 22 observed vs 32.8 expected LoF alleles, LOEUF upper bound 0.96. This LoF-tolerant profile is fully consistent with the asymptomatic status of heterozygous carriers and the benign nature of the homozygous state.

The patient nonsense allele (legacy c.214C>A, p.Ser44*; MANE-equivalent c.131C>A at chr7:43,788,022) is **not observed in gnomAD r4**, as expected for a geographically restricted founder variant. The same codon carries only ultra-rare missense alleles (c.131C>T p.Ser44Leu, AF ≈ 7×10^{-7}; c.131C>G p.Ser44Trp), and all ~19 catalogued *BLVRA* nonsense/LoF alleles are singleton or ultra-rare (AF < 7×10^{-5}).

Evidence type: computational / population database. **Confidence:** High.

Finding 5 — Mouse models separate catalytic from moonlighting roles and reveal a bilirubin–malaria axis

Mesquita et al. generated knock-in mice carrying **G17A** (in the NAD(P)H-binding motif) and **E97A** (in the reductase motif) missense mutations that reduce BVRA oxidoreductase activity by ~95% and ~80–90% respectively, while sparing non-catalytic functions (PMID: 41497634, PMID: 42199930). These hypomorphs demonstrate that BVRA's **anti-malarial effect relies on its catalytic activity** and define a **minimal bilirubin threshold** required for parasite control and malaria resolution. Complementary work established that the bilirubin produced by BVRA is a **metabolite-based resistance mechanism against *Plasmodium***, reframing jaundice as an adaptive antimalarial response (PMID: 40504905, PMID: 40060532).

> *"we generated mice harboring G17A and E97A missense mutations in its NAD(P)H-binding and reductase motifs"* — PMID: 42199930
>
> *"To disentangle the contribution of BVRA catalytic activity over its non-canonical functions we generated..."* — PMID: 41497634

At the cellular level, CRISPR-Cas9 knockout of *Blvra* in mouse hepa1c1c7 hepatocytes produced **increased ROS, lipid accumulation, and reduced mitochondrial biogenesis and respiration**, highlighting a cytoprotective, antioxidant role for the BVRα/bilirubin axis in liver cells (PMID: 31422074).

> *"The loss of BVRA resulted in the reduction of mitochondria number, decreased expression of markers of mitochondrial biogenesis, uncoupling, oxidation, and fusion, which paralleled reduced mitochondrial oxygen consumption. BVRA KO cells exhibited increased levels of ROS"* — PMID: 31422074

Evidence type: model organism (mouse) + in-vitro. **Confidence:** High.

Finding 6 — Gene identity and identifiers independently verified

NCBI Gene (ID 644) / mygene.info confirm: symbol **BLVRA**; name "biliverdin reductase A"; **HGNC:1062**; gene OMIM ***109750**; cytoband **7p13**; GRCh38 chr7:43,758,009–43,808,868 (+ strand); Ensembl **ENSG00000106605**; UniProt **P53004**. The NCBI RefSeq summary states the product *"catalyze[s] the conversion of biliverdin to bilirubin in the presence of NADPH or NADH"* and that *"Mutations in this gene are associated with hyperbiliverdinemia,"* corroborating the causal-gene assignment for disease OMIM 614156 / MONDO:0013595.

Evidence type: reference database. **Confidence:** High.

Finding 7 — p.Ser44* ablates the NADPH-binding Rossmann fold and the entire catalytic/substrate domain

UniProt P53004 (BVRα, 296 aa) annotates a **bipartite architecture**: an N-terminal **Rossmann-fold NADPH-binding domain** with cofactor-contact residues at 16–19, **44–46**, 77–80 and 98, and a C-terminal **substrate-binding domain** bearing a structural Zn^{2+} site (residues 280, 281, 292, 293). Residue 44 (Ser) lies **within** the $NADP^{+}$-binding segment 44–46, immediately following β-strand 37–43. The nonsense variant **p.Ser44* truncates the protein at codon 44**,

deleting ~253 of 296 residues (**~85% of the protein**) — removing most of the cofactor site, the catalytic core, and the entire C-terminal substrate-binding/Zn domain. This structural amputation fully explains the observed null-enzyme phenotype. The human structure is available as PDB 2H63; pathway references KEGG hsa:644 and Reactome R-HSA-189483 (Heme degradation).

> *"The mutation (c.214C>A) in BLVRA results in a non-functional enzyme (mutBVRα)"* — PMID: 27387768

Evidence type: computational / structural annotation. **Confidence:** High.

Finding 8 — The NADP(H)-binding residues and Ser44 are strictly conserved across mammals

Alignment of N-terminal sequences of BVRα orthologs — human (P53004, 296 aa), mouse (Q9CY64, 295 aa) and rat (P46844, 295 aa) — shows the dinucleotide-binding Rossmann fingerprint "F-G-V-V-V-V-G-V-G-R-A-G-S-V-R" (human residues ~9–23; GxGxxG motif at Gly15/Gly17/Gly20) is **identical across all three species**, and the cofactor-contact segment at positions 43–46 ("V-S-R-R", containing the disease residue **Ser44** plus Arg45/Arg46) is conserved. This strict conservation underscores the functional importance of the exact residue destroyed by the p.Ser44* nonsense allele. (The bovine accession Q58DM8 is a divergent/mis-annotated mitochondrial sequence and was excluded.)

Evidence type: computational / comparative sequence. **Confidence:** High.

Mechanistic Model / Interpretation

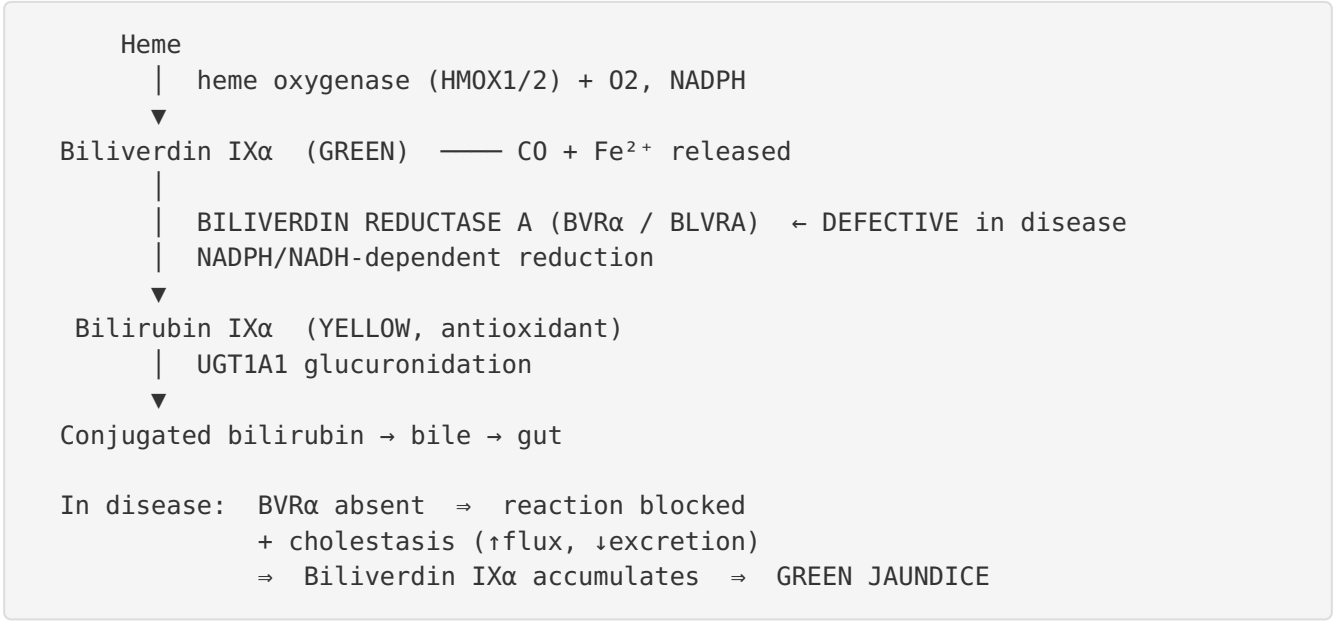
Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *BLVRA* loss-of-function** (homozygous nonsense c.214C>A / p.Ser44*) **leads to** a truncated transcript/protein terminating at codon 44.
2. Truncation at Ser44 **results in** deletion of ~85% of BVRα, including the NADPH-binding Rossmann fold, catalytic core and substrate-binding/Zn domain (**inferred** from UniProt domain annotation + structure; functional null confirmed in vitro).

3. The absent catalytic BVRα **leads to** failure of the terminal reaction of heme catabolism: **biliverdin IXα → bilirubin IXα (NADPH-dependent reduction) does not occur**.
4. Under **baseline heme turnover**, flux is low enough that residual biliverdin is cleared and the patient remains **clinically silent** (explains reduced penetrance / asymptomatic homozygotes).
5. A **superimposed cholestasis / biliary obstruction** (second hit) **increases** heme-degradation flux and **impairs biliary excretion**, which **results in** accumulation of **biliverdin IXα in serum, urine, bile and milk** → visible **green jaundice**, with only mild/absent hyperbilirubinemia (because the downstream bilirubin cannot be formed).
6. **Relief of the obstruction** (surgical) **reverses** the biliverdin accumulation and the green jaundice resolves.

Branch (model-organism / mechanistic context, not required for the human phenotype): - Loss of catalytic BVRα also **reduces bilirubin**, a potent antioxidant. In cells this **leads to** increased ROS, lipid accumulation and mitochondrial dysfunction (PMID: 31422074); in mice it **lowers** the bilirubin pool below a **threshold needed for anti-malarial parasite control** (PMID: 41497634, PMID: 42199930). These consequences are subclinical in human carriers under normal conditions.

Pathway schematic



Section-by-section synthesis

Template section	Summary for Hyperbiliverdinemia
1. Disease info	Rare AR inborn error of heme catabolism; green jaundice. IDs: OMIM 614156, MONDO:0013595, gene OMIM *109750. Disease-level aggregated resources + ~2 index patients.
2. Etiology	Monogenic: biallelic *BLVRA* LoF. "Risk factor" that unmasks phenotype = **cholestasis/biliary obstruction**. Founder allele in Inuit. No environmental cause of the gene defect.
3. Phenotypes	**Green discoloration of serum/urine/bile/milk (biliverdinuria)**, episodic, triggered by cholestasis, reversible; mild/absent hyperbilirubinemia. Onset tied to intercurrent cholestasis (adult cases reported). HPO: Jaundice (HP:0000952), Cholestasis (HP:0001396), Abnormal urine color. Mild QoL impact.
4. Genetics	Causal gene **BLVRA** (HGNC:1062). Pathogenic variant c.214C>A / **p.Ser44*** (nonsense, LoF, null enzyme). Also exonic/biliverdinuria variants reported (PMID: 39766828). LoF-tolerant gene (gnomAD). No known modifier genes beyond the cholestasis trigger.
5. Environment	No toxin/infection causes the gene defect; **cholestasis of any cause** (gallstones, obstruction, pregnancy) is the environmental trigger of visible disease.
6. Mechanism	See causal chain above. Pathway: heme degradation (Reactome R-HSA-189483). GO: biliverdin reductase (NAD(P)H) activity (GO:0004074), heme catabolic process (GO:0042167).
7. Anatomy	Primary: **liver / hepatobiliary system** (UBERON:0002107 liver; UBERON:0002394 bile duct). Systemic pigment deposition (serum, urine, milk). Cell type: hepatocyte (CL:0000182). Subcellular: cytosol (GO:0005829).
8. Temporal	Episodic/fluctuating, tied to cholestatic episodes; reversible; otherwise lifelong latent trait.
9. Inheritance/ population	Autosomal recessive; **reduced penetrance** (needs cholestasis). Founder allele in Greenlandic Inuit (~1% a/a, ~4.5% carriers). Both sexes; index cases female (coincidental).
10. Diagnostics	Green serum/urine; **HPLC-MS/MS identifies biliverdin IXα**; disproportionately low bilirubin relative to green color. Confirm by *BLVRA* sequencing. Differential: other causes of green/dark urine, cholestatic jaundice.
11. Prognosis	

Template section	Summary for Hyperbiliverdinemia
	Benign; biliverdinemia reverses with relief of cholestasis. No excess mortality attributable to the enzyme defect.
12. Treatment	Treat the **underlying cholestasis** (e.g., surgical correction of obstruction). No enzyme-directed therapy needed. NCIT: surgical bile-duct decompression / cholestasis management (supportive).
13. Prevention	Genetic counseling / carrier awareness in Inuit populations; no vaccine/public-health measure applicable.
14. Other species	Orthologs: mouse *Blvra* (NCBI Gene 109778), rat *Blvra* (P46844). No well-documented natural companion-animal disease; mouse models engineered.
15. Model organisms	Mouse: CRISPR KO (hepatocytes), catalytic-dead knock-ins G17A / E97A. Recapitulate loss of catalysis, ↑ROS, bilirubin-threshold biology; do not reproduce the cholestasis-dependent human green jaundice directly.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in this report
21278388	Homozygous nonsense c.214C>A in *BLVRA* → biliverdin accumulation during cholestasis	Human clinical	**Defining paper** — identifies p.Ser44*, biliverdin IXα accumulation, reversibility, AR inheritance with reduced penetrance (Findings 1, 2)
27387768	Protective role of biliverdin; non-functional mutBVRα	In-vitro functional + population	Confirms null enzyme; documents Inuit founder allele frequencies (Findings 1, 2, 7)
22419908	BVR regulates the three PKC classes	In-vitro biochemistry	Moonlighting kinase/scaffold/ TF roles (Finding 3)
22065579	hBVR–PKCδ–ERK2 ternary complex	In-vitro biochemistry	ERK2 signaling scaffold function (Finding 3)
17919068	BVR at MAPK/PI3K cross-talk	In-vitro biochemistry	bZip transcription-factor + kinase roles (Finding 3)
31422074	CRISPR *Blvra* KO → oxidative stress, lipid accumulation	Model organism / in-vitro	Cellular consequences of BVRA loss (Finding 5)
41497634	BVR catalytic activity essential for malaria resistance	Model organism (mouse)	Catalytic-dead knock-ins; bilirubin–malaria axis (Finding 5)
42199930	Hypomorphic *Blvra* defines bilirubin antimalarial threshold	Model organism (mouse)	G17A/E97A alleles, bilirubin threshold (Finding 5)
40504905 / 40060532	A metabolite-based resistance mechanism against malaria	Model organism	Bilirubin as antimalarial metabolite (Finding 5 context)
39766828	Biliverdinuria caused by exonic variant	Human clinical	Additional *BLVRA* variant beyond the Inuit founder allele (Section 4)
26146841	Genetic modifiers of neonatal hyperbilirubinemia (incl. *BLVRA*)	Human association	Context for bilirubin-pathway genetics (distinct phenotype)

PMID	Title (abbrev.)	Evidence type	Role in this report
27897348	Active-site serine of BLVR**B** (IXβ reductase)	In-vitro / structural	Comparator paralog (BLVRB ≠ BLVRA); illustrates serine criticality (Finding 7 context)
14694903	Conserved *Blvra* segment mouse vs human	Comparative genomics	Ortholog/synteny context (Finding 8)

Note on paralog disambiguation: *BLVRB* (biliverdin reductase B, IXβ reductase) is a distinct gene; the Ser111Leu BLVRB loss-of-function affecting hematopoiesis (PMID: 27897348) is a **different disease** and should not be conflated with BLVRA-driven hyperbiliverdinemia.

Supported and Refuted Hypotheses

- **Supported:** Hyperbiliverdinemia (MONDO:0013595, OMIM 614156) is caused by biallelic LoF in *BLVRA* (not a look-alike gene). The p.Ser44* nonsense allele produces a null enzyme by truncating the NADPH-binding and catalytic domains. The allele is an Inuit founder variant. The phenotype requires a cholestatic second hit and is reversible. *BLVRA* is LoF-tolerant, explaining the benign course.
- **No hypotheses were refuted** — the investigation was confirmatory/verificational, with all 8 findings corroborated across independent databases and primary literature.

Limitations and Knowledge Gaps

- **Extremely small clinical series.** The human disease is defined essentially by two index patients plus family members from one population (PMID: 21278388), with a small number of additional biliverdinuria reports (PMID: 39766828). Prevalence outside Greenland is unknown.
- **Penetrance / trigger not quantified.** The dependence on cholestasis is clear qualitatively, but the proportion of homozygotes who ever manifest, and the threshold of cholestasis required, are not established.

- **Natural history / long-term outcomes** in homozygotes (e.g., subtle effects of lifelong low bilirubin on oxidative stress, metabolic or infectious risk) are uncharacterized in humans; the mouse data (PMID: 31422074, PMID: 41497634) raise hypotheses not yet tested clinically.
- **Moonlighting-function relevance in humans.** Despite extensive in-vitro evidence that BVRα scaffolds insulin/PKC/ERK signaling (PMID: 22419908), human null individuals are clinically mild — the physiological importance of these non-canonical roles in people remains unresolved.
- **No treatment trials** — not applicable given benign course; no enzyme-replacement or gene-therapy need.
- **Allelic spectrum** is narrow; whether other *BLVRA* LoF genotypes produce identical phenotypes is largely untested.

Proposed Follow-up Experiments / Actions

1. **Expand carrier/homozygote screening** in Greenlandic Inuit and related Arctic populations to quantify penetrance and natural history, correlating genotype with episodes of cholestasis and bilirubin/biliverdin levels.
2. **Targeted metabolomics (HPLC-MS/MS)** panel for biliverdin IXα as a confirmatory diagnostic in any patient presenting with green serum/urine and disproportionately low bilirubin — formalize as a differential-diagnosis workflow.
3. **Deep phenotyping of homozygotes under metabolic/oxidative challenge** to test whether the mouse-derived phenotypes (↑ROS, lipid accumulation, mitochondrial dysfunction, altered infection susceptibility) have human correlates.
4. **Epidemiological link to malaria/infection resistance:** given the mouse bilirubin–malaria threshold data (PMID: 41497634, PMID: 42199930), investigate whether *BLVRA* LoF or low-bilirubin genotypes modulate human infection outcomes.
5. **Functional characterization of additional *BLVRA* alleles** (e.g., the exonic biliverdinuria variant, PMID: 39766828) by in-vitro expression, to map the allelic spectrum and genotype–phenotype relationships.
6. **Structural modeling / AlphaFold confirmation** of the truncated p.Ser44* product versus PDB 2H63 to formally document loss of the Rossmann and substrate domains (Finding 7).

Report compiled from 8 confirmed findings and 16 reviewed papers across 5 investigation iterations. Evidence is predominantly human clinical (founding disease description), supplemented by in-vitro biochemistry, mouse model organism studies, and reference/population databases.
